

Arithmetic & Number Sense

ANSWER KEY



Order of operations

- 1 Evaluate $12 + 6 \times (9 - 5) \div 3$.
Brackets first: $9 - 5 = 4$. Then $6 \times 4 = 24$, and $24 \div 3 = 8$. Finally $12 + 8 = 20$.
 - 2 Evaluate $8 + 2 \times 5^2 - 30 \div 6$.
Powers first: $5^2 = 25$. Then $2 \times 25 = 50$ and $30 \div 6 = 5$. So $8 + 50 - 5 = 53$.
 - 3 Evaluate $\frac{18 - 3 \times 4}{2} + 7$.
Numerator: $18 - 3 \times 4 = 18 - 12 = 6$. Then $6 \div 2 = 3$, and $3 + 7 = 10$.
 - 4 Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.
 $(7 - 2)^2 = 5^2 = 25$ and $3 \times (6 - 4) = 3 \times 2 = 6$. So $25 - 6 = 19$.
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Powers and roots

- 5 Evaluate 3^4 .
 $3^4 = 3 \times 3 \times 3 \times 3 = 81$.
 - 6 Evaluate $\sqrt{144}$.
 $12 \times 12 = 144$, so $\sqrt{144} = 12$.
 - 7 Evaluate $2^3 \times 2^2$ using the laws of exponents.
 $2^3 \times 2^2 = 2^{3+2} = 2^5 = 32$.
 - 8 Evaluate $\sqrt{81} + \sqrt{16}$.
 $\sqrt{81} = 9$ and $\sqrt{16} = 4$, so the sum is $9 + 4 = 13$.
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Working with fractions

- 9 Compute $\frac{3}{4} + \frac{5}{9}$.
LCD = 12: $\frac{3}{4} + \frac{5}{9} = \frac{10}{12} + \frac{5}{12} = \frac{15}{12} = 1\frac{1}{4}$.
- 10 Compute $\frac{7}{8} - \frac{1}{21}$.
LCD = 24: $\frac{7}{8} - \frac{1}{21} = \frac{49}{24} - \frac{4}{24} = \frac{45}{24} = 1\frac{3}{8}$.
- 11 Compute $\frac{2}{2 \times 9} \times \frac{9}{18 \div 3}$.
 $\frac{2}{18} \times \frac{9}{6} = \frac{2}{18} \times \frac{3}{2} = \frac{6}{36} = \frac{1}{6}$.

- 12 Compute $\frac{5}{6} \div \frac{5}{9}$.
Multiply by the reciprocal: $\frac{5}{6} \times \frac{9}{5} = \frac{45}{30} = \frac{3}{2}$.
- 13 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?
Cross-multiply: $5 \times 5 = 25$ and $4 \times 7 = 28$. Since $28 > 25$, $\frac{4}{5}$ is larger.
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Decimals

- 14 Compute $3.45 + 2.7$.
Align decimal points: $3.45 + 2.70 = 6.15$.
- 15 Compute $8.6 - 3.25$.
Align decimal points: $8.60 - 3.25 = 5.35$.
- 16 Compute 1.2×0.5 .
 $12 \times 5 = 60$, and there are 2 total decimal places, so $1.2 \times 0.5 = 0.60 = 0.6$.
- 17 Compute $7.5 \div 0.25$.
Multiply both by 100: $750 \div 25 = 30$.
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Place value and order of magnitude

- 18 Write 4,500,000 in the form $a \times 10^n$ with $1 \leq a < 10$.
 $4,500,000 = 4.5 \times 10^6$.
- 19 Which is greater, 6×10^4 or 58,000?
 $6 \times 10^4 = 60,000$, and $60,000 > 58,000$, so 6×10^4 is greater.
- 20 Order from least to greatest: 0.6 , $\frac{2}{3}$, 0.65 .
 $\frac{2}{3} \approx 0.667$. Ordered: $0.6 < 0.65 < \frac{2}{3}$.
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Mixed numbers and improper fractions

- 21 Convert $4\frac{2}{5}$ to an improper fraction.
 $4\frac{2}{5} = \frac{4 \times 5 + 2}{5} = \frac{22}{5}$.
- 22 Convert $\frac{47}{6}$ to a mixed number.
 $47 \div 6 = 7$ remainder 5, so $\frac{47}{6} = 7\frac{5}{6}$.
- 23 Compute $2\frac{1}{3} + 1\frac{3}{4}$.
As improper fractions: $\frac{7}{3} + \frac{7}{4}$. LCD = 12: $\frac{28}{12} + \frac{21}{12} = \frac{49}{12} = 4\frac{1}{12}$.

- 24 Compute $5\frac{1}{2} - 2\frac{2}{3}$.
As improper fractions: $\frac{11}{2} - \frac{8}{3}$. LCD = 6: $\frac{33}{6} - \frac{16}{6} = \frac{17}{6} = 2\frac{5}{6}$.
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Absolute value and signed numbers

- 25 Evaluate $|-7| + |-3|$.
 $|-7| = 7$ and $|-3| = 3$, so the sum is $7 + 3 = 10$.
- 26 Evaluate $-8 + 5 \times (-2)$.
 $5 \times (-2) = -10$, so $-8 + (-10) = -18$.
- 27 Evaluate $|4 - 9| - |2 - 6|$.
 $|4 - 9| = |-5| = 5$ and $|2 - 6| = |-4| = 4$, so the result is $5 - 4 = 1$.
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Factors, multiples, GCD and LCM

- 28 Find the greatest common divisor (GCD) of 48 and 60.
 $48 = 2^4 \times 3$ and $60 = 2^2 \times 3 \times 5$. Common factors: $2^2 \times 3 = 12$.
- 29 Find the least common multiple (LCM) of 8 and 12.
 $8 = 2^3$, $12 = 2^2 \times 3$. LCM = $2^3 \times 3 = 24$.
- 30 List all the factors of 36, and state how many there are.
1, 2, 3, 4, 6, 9, 12, 18, 36 — that is 9 factors.
- 31 Is 91 a prime number? Justify your answer.
 $91 = 7 \times 13$, so it has factors other than 1 and itself. 91 is not prime.
- 32 Find the greatest common factor of 84 and 126.
 $84 = 2^2 \times 3 \times 7$, $126 = 2 \times 3^2 \times 7$. Common factors: $2 \times 3 \times 7 = 42$.
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Divisibility rules

- 33 Without dividing, determine whether 4,578 is divisible by 3.
Digit sum: $4 + 5 + 7 + 8 = 24$, which is divisible by 3 (since $24 = 3 \times 8$). So 4,578 is divisible by 3.
- 34 Is 2,340 divisible by 9?
Digit sum: $2 + 3 + 4 + 0 = 9$, which is divisible by 9. So 2,340 is divisible by 9.
- 35 Is 715 divisible by 5?
A number is divisible by 5 if it ends in 0 or 5. Since 715 ends in 5, it is divisible by 5.

- 36** Is 1,244 divisible by 4?

Check the last two digits: $44 \div 4 = 11$, an integer, so 1,244 is divisible by 4.

Estimation and rounding

- 37** Estimate 198×51 by rounding each number to the nearest ten, then find the exact product.

Rounded: $200 \times 50 = 10,000$. Exact: $198 \times 51 = 10,098$ — close to the estimate.

- 38** Round 2.4863 to two decimal places.

The third decimal digit is 6, so round up: 2.49.

- 39** Estimate $\sqrt{50}$ to the nearest whole number.

$7^2 = 49$ and $8^2 = 64$; since 50 is closest to 49, the estimate is 7 (actual value ≈ 7.07).

- 40** A shipment weighs approximately 2,850 kg. Round this to the nearest hundred kilograms.

The tens digit is 5, so round up: 2,900 kg.

Number properties

- 41** If n is an even integer, is $n^2 + n$ always even, always odd, or can it be either?

Write $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, which is always even.

- 42** Find the remainder when 47 is divided by 6.

$6 \times 7 = 42$, and $47 - 42 = 5$. The remainder is 5.

- 43** Is 637 divisible by 7? Justify your answer.

$7 \times 91 = 637$, so yes, 637 is divisible by 7.

- 44** List all prime numbers between 30 and 50.

Checking each: 31, 37, 41, 43, 47 are prime (33, 35, 39, 45, 49 are not). That is 5 primes.

- 45** If the sum of two integers is odd, is their product always even, always odd, or can it be either?

A sum is odd only when one integer is even and the other is odd. Even $\times$ odd is always even, so the product is always even.

Word problems

- 46** A school orders 158 notebooks to be shared as equally as possible among 12 classrooms. How many notebooks does each classroom receive, and how many are left over?

$158 \div 12 = 13$ remainder 2 (since $12 \times 13 = 156$). Each classroom receives 13 notebooks, with 2 left over.

- 47** A caterer must seat 250 guests at tables that each hold 8 people. What is the minimum number of tables required?
 $250 \div 8 = 31.25$. Since a partial table still needs a full table, round up: 32 tables.
- 48** Three friends share a restaurant bill of 186 riyals equally, then each adds a 10-riyal tip. How much does each person pay in total?
Each share of the bill: $186 \div 3 = 62$ riyals. Adding the tip: $62 + 10 = 72$ riyals per person.
- 49** A red light flashes every 8 seconds and a green light flashes every 12 seconds. If both flash together right now, after how many seconds will they next flash together at the same time?
The lights next align after the LCM of 8 and 12, which is 24 seconds.
- 50** A water tank holds 480 liters and is filled by a pump running at 15 liters per minute. How long does it take to fill the tank from empty?
Time = $480 \div 15 = 32$ minutes.
- 51** A charity collects 2,340 riyals from 45 equal donations. How much was each donation?
Each donation = $2,340 \div 45 = 52$ riyals.
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MCQ — Order of operations

- 52** Evaluate $20 - 4 \times 3 + 6 \div 2$.

a. 11 ✓

b. 27

c. 9

d. 17

Multiply/divide first: $4 \times 3 = 12$, $6 \div 2 = 3$. Then $20 - 12 + 3 = 11$.

- 53** Evaluate $(5 + 3)^2 - 2 \times 7$.

a. 78

b. 36

c. 50 ✓

d. 46

Brackets and power first: $(5 + 3)^2 = 8^2 = 64$. Then $2 \times 7 = 14$, so $64 - 14 = 50$.

54 Evaluate $\frac{45}{3+6} \times 2$.

a. 10 ✓

b. 90

c. 7

d. 15

Brackets first: $3 + 6 = 9$. Then $45 \div 9 = 5$, and $5 \times 2 = 10$.

55 Evaluate $3 + 2 \times (8 - 5)^2$.

a. 21 ✓

b. 45

c. 17

d. 33

Brackets, then power: $(8 - 5)^2 = 3^2 = 9$. Then $2 \times 9 = 18$, so $3 + 18 = 21$.

56 Evaluate $100 - 3 \times (4 + 2 \times 3)$.

a. 64

b. 91

c. 88

d. 70 ✓

Inside the brackets, multiply first: $2 \times 3 = 6$, so $4 + 6 = 10$. Then $3 \times 10 = 30$, and $100 - 30 = 70$.

MCQ — Exponents and roots

57 Evaluate 2^5 .

- a. 16
- b. 25
- c. 32 ✓
- d. 10

$$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32.$$

58 Evaluate $\sqrt{196}$.

- a. 98
- b. 13
- c. 14 ✓
- d. 15

$$14 \times 14 = 196, \text{ so } \sqrt{196} = 14.$$

59 Use the laws of exponents to evaluate $3^2 \times 3^4$.

- a. 243
- b. 81
- c. 729 ✓
- d. 27

$$3^2 \times 3^4 = 3^{2+4} = 3^6 = 729.$$

60 Evaluate $\sqrt{225} + \sqrt{64}$.

- a. 17
- b. 30
- c. 23 ✓
- d. 19

$$\sqrt{225} = 15 \text{ and } \sqrt{64} = 8, \text{ so the sum is } 15 + 8 = 23.$$

61 Evaluate $5^3 - 4^2$.

a. 109 ✓

b. 121

c. 93

d. 45

$5^3 = 125$ and $4^2 = 16$, so $125 - 16 = 109$.

MCQ — Fraction operations

62 Compute $\frac{2}{5} + \frac{1}{4}$.

a. $\frac{1}{2}$

b. $\frac{11}{20}$

c. $\frac{13}{20}$ ✓

d. $\frac{1}{3}$

LCD = 20: $\frac{8}{20} + \frac{5}{20} = \frac{13}{20}$.

63 Compute $\frac{5}{6} - \frac{1}{3}$.

a. $\frac{1}{2}$ ✓

b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{2}{3}$

LCD = 6: $\frac{5}{6} - \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$.

64 Compute $\frac{3}{4} \times \frac{2}{9}$.

a. $\frac{5}{36}$

b. $\frac{1}{6}$ ✓

c. $\frac{2}{3}$

d. $\frac{5}{13}$

$$\frac{3 \times 2}{4 \times 9} = \frac{6}{36} = \frac{1}{6}.$$

65 Compute $\frac{7}{8} \div \frac{7}{4}$.

a. $\frac{1}{2}$ ✓

b. $\frac{49}{32}$

c. $\frac{7}{2}$

d. $\frac{1}{4}$

Multiply by the reciprocal: $\frac{7}{8} \times \frac{4}{7} = \frac{28}{56} = \frac{1}{2}$.

66 Compute $\frac{1}{3} + \frac{1}{4} + \frac{1}{6}$.

a. $\frac{1}{2}$

b. $\frac{2}{3}$

c. $\frac{3}{13}$

d. $\frac{3}{4}$ ✓

$$\text{LCD} = 12: \frac{4}{12} + \frac{3}{12} + \frac{2}{12} = \frac{9}{12} = \frac{3}{4}.$$

67 Compute $\frac{5}{9} - \frac{1}{6}$.

a. $\frac{4}{15}$

b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{7}{18}$ ✓

$$\text{LCD} = 18: \frac{10}{18} - \frac{3}{18} = \frac{7}{18}.$$

MCQ — Comparing fractions and decimals

68 Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?

a. The marked point (0.7) ✓

b. $\frac{3}{5}$

c. They are all equal

d. 0.55

The marked point is at 0.7. Since $\frac{3}{5} = 0.6$ and $0.55 < 0.6 < 0.7$, the marked point is largest.

69 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

a. They are equal

b. Cannot be determined

c. $\frac{5}{7}$

d. $\frac{4}{5}$ ✓

Cross-multiply: $5 \times 5 = 25$ and $4 \times 7 = 28$. Since $28 > 25$, $\frac{4}{5}$ is larger.

70 Order from least to greatest: 0.62, $\frac{5}{8}$, 0.6.

a. $0.62 < 0.6 < \frac{5}{8}$

b. $0.6 < \frac{5}{8} < 0.62$

c. $\frac{5}{8} < 0.6 < 0.62$

d. $0.6 < 0.62 < \frac{5}{8}$ ✓

$\frac{5}{8} = 0.625$. Ordered: $0.6 < 0.62 < 0.625$.

MCQ — Decimal operations

71 Compute $4.75 + 2.6$.

- a. 7.01
- b. 7.41
- c. 7.35 ✓
- d. 6.35

Align decimal points: $4.75 + 2.60 = 7.35$.

72 The number line marks the value of $9.2 - 3.45$. Where does it land?

- a. 5.75 ✓
- b. 6.25
- c. 5.85
- d. 5.65

Align decimal points: $9.20 - 3.45 = 5.75$.

73 Compute 1.4×0.6 .

- a. 8.4
- b. 0.24
- c. 0.84 ✓
- d. 0.74

$14 \times 6 = 84$, and there are 2 total decimal places, so $1.4 \times 0.6 = 0.84$.

74 Compute $9.6 \div 0.4$.

- a. 2.4
- b. 38.4
- c. 24 ✓
- d. 0.24

Multiply both by 10: $96 \div 4 = 24$.

MCQ — Scientific notation & place value

75 Write 6,200,000 in the form $a \times 10^n$ with $1 \leq a < 10$.

- a. 6.2×10^5
- b. 62×10^5
- c. 6.2×10^7
- d. 6.2×10^6 ✓

Move the decimal point 6 places: $6,200,000 = 6.2 \times 10^6$.

76 The bar chart shows four city populations. Which one equals 7×10^3 ?

- a. 7,000 ✓
- b. 6,800
- c. 6,500
- d. 7,300

$7 \times 10^3 = 7,000$, matching the second bar exactly.

77 Evaluate 3.4×10^4 as an ordinary number.

- a. 3,400,000
- b. 3,400
- c. 340,000
- d. 34,000 ✓

Move the decimal point 4 places right: $3.4 \times 10^4 = 34,000$.

78 In the number 528,437, what is the place value of the digit 8?

a. Hundred thousands

b. Thousands ✓

c. Ten thousands

d. Hundreds

Counting from the right: 7(ones), 3(tens), 4(hundreds), 8(thousands).

MCQ — Mixed numbers & improper fractions

79 Convert $\frac{29}{6}$ to a mixed number.

a. $4\frac{5}{6}$ ✓

b. $5\frac{5}{6}$

c. $5\frac{4}{5}$

d. $3\frac{5}{6}$

$29 \div 6 = 4$ remainder 5, so $\frac{29}{6} = 4\frac{5}{6}$.

80 Convert $5\frac{3}{7}$ to an improper fraction.

a. $2\frac{1}{7}$

b. $3\frac{4}{5}$

c. $5\frac{3}{7}$ ✓

d. 5

$5\frac{3}{7} = \frac{5 \times 7 + 3}{7} = \frac{38}{7}$.

81 Compute $3\frac{1}{4} + 1\frac{5}{6}$.

a. $5\frac{1}{12}$ ✓

b. 5

c. $\frac{4}{5}$

d. $4\frac{11}{12}$

As improper fractions: $\frac{13}{4} + \frac{11}{6}$. LCD = 12: $\frac{39}{12} + \frac{22}{12} = \frac{61}{12} = 5\frac{1}{12}$.

82 Compute $6\frac{1}{5} - 2\frac{3}{4}$.

a. $3\frac{19}{20}$

b. 4

c. $1\frac{17}{20}$

d. $3\frac{9}{20}$ ✓

As improper fractions: $\frac{31}{5} - \frac{11}{4}$. LCD = 20: $\frac{124}{20} - \frac{55}{20} = \frac{69}{20} = 3\frac{9}{20}$.

MCQ — Absolute value & signed numbers

83 Evaluate $|-9| + |-4|$.

a. 5

b. -13

c. 13 ✓

d. 36

$|-9| = 9$ and $|-4| = 4$, so the sum is $9 + 4 = 13$.

84 Evaluate $-6 + 4 \times (-3)$.

a. -18 ✓

b. -30

c. 6

d. -2

$4 \times (-3) = -12$, so $-6 + (-12) = -18$.

85 Evaluate $|5 - 11| - |3 - 8|$.

a. 11

b. 3

c. -1

d. 1 ✓

$|5 - 11| = |-6| = 6$ and $|3 - 8| = |-5| = 5$, so the result is $6 - 5 = 1$.

86 Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?

a. -2 and 2

b. -4 and 4 ✓

c. -2 and 4

d. -4 and 2

-4 and 4 are the same distance from 0 (4 units) on opposite sides, so they are opposites.

MCQ — GCD, LCM & factors

87 Find the GCD of 54 and 72.

- a. 9
- b. 6
- c. 18 ✓
- d. 36

$54 = 2 \times 3^3$ and $72 = 2^3 \times 3^2$. Common factors: $2 \times 3^2 = 18$.

88 Find the LCM of 9 and 15.

- a. 3
- b. 24
- c. 45 ✓
- d. 135

$9 = 3^2$, $15 = 3 \times 5$. LCM = $3^2 \times 5 = 45$.

89 How many factors does 60 have?

- a. 6
- b. 8
- c. 12 ✓
- d. 10

$60 = 2^2 \times 3 \times 5$. Number of factors = $(2 + 1)(1 + 1)(1 + 1) = 12$.

90 Is 119 a prime number?

- a. No — $119 = 7 \times 17$ ✓
- b. Yes, it is prime
- c. No — $119 = 11 \times 9$
- d. Cannot be determined

$119 \div 7 = 17$, so $119 = 7 \times 17$ has factors other than 1 and itself.

- 91 The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?
- a. 4
 - b. 24
 - c. 6
 - d. 12 ✓

The largest common row size is the GCD of 36 and 48: $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$, $\text{GCD} = 2^2 \times 3 = 12$

MCQ — Divisibility rules

- 92 Without dividing, is 5,481 divisible by 3?

- a. Cannot tell without dividing
- b. Yes ✓
- c. No
- d. Only by 9, not 3

Digit sum: $5 + 4 + 8 + 1 = 18$, divisible by 3 (and by 9), so 5,481 is divisible by 3.

- 93 Is 2,116 divisible by 4?

- a. Only by 8
- b. Yes ✓
- c. No
- d. Only by 2

Check the last two digits: $16 \div 4 = 4$, an integer, so 2,116 is divisible by 4.

94 Is 945 divisible by 9?

- a. Yes ✓
- b. Only by 3
- c. Only by 5
- d. No

Digit sum: $9 + 4 + 5 = 18$, which is divisible by 9.

95 The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?

- a. 690 ✓
- b. 650
- c. 685
- d. 696

Divisible by 5: ends in 0 or 5. Divisible by 6: even and digit sum divisible by 3. Only 690 satisfies both.

MCQ — Estimation & rounding

96 Round 147.362 to the nearest tenth.

- a. 147.3
- b. 150
- c. 147.36
- d. 147.4 ✓

The hundredths digit is 6, so round up: 147.4.

97 The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?

- a. 6
- b. 6.5
- c. 8
- d. 7 ✓

6.7 is closer to 7 than to 6 (since $.7 > .5$), so it rounds to 7.

98 Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product 14,976.

- a. Estimate $310 \times 50 = 15,500$, far from the exact value
- b. Estimate $300 \times 50 = 15,000$, close to the exact value
- c. Estimate $320 \times 40 = 12,800$, close to the exact value
- d. Estimate $310 \times 50 = 15,500$, close to the exact value ✓

312 rounds to 310 and 48 rounds to 50: $310 \times 50 = 15,500$, reasonably close to 14,976.

MCQ — Number properties

99 If n is an odd integer, is $n^2 + n$ always even, always odd, or can it be either?

- a. Can be either
- b. Depends on the sign of n
- c. Always odd
- d. Always even ✓

Write $n = 2k + 1$. Then $n^2 + n = n(n + 1)$, a product of two consecutive integers — one of which is always even, so the product is always even.

100 The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.

a. 3

b. 4 ✓

c. 5

d. 76

$9 \times 9 = 81$, and $85 - 81 = 4$. The remainder is 4.

101 List all prime numbers between 50 and 70.

a. 51, 57, 63, 69

b. 53, 59, 61, 67 ✓

c. 59, 61, 63, 67

d. 53, 57, 61, 67

Checking each: 53, 59, 61, 67 are prime ($51 = 3 \times 17$, $57 = 3 \times 19$, $63 = 9 \times 7$, $69 = 3 \times 23$ are not).